

Understanding the Kinetic Theory of Gases

Learn What is Molecular Motion and Gas Behaviour?

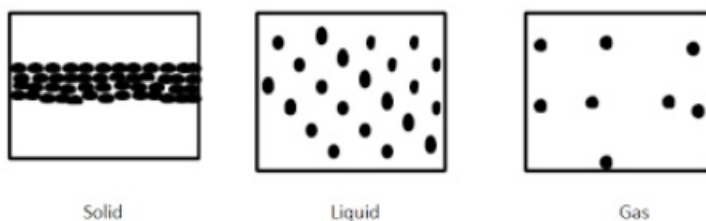
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The Kinetic Theory of Gases explains gas behaviour by treating molecules as constantly moving particles. It helps derive laws like Boyle's Law and Charles's Law, showing how pressure, volume, and temperature are related. The theory assumes random molecular motion, elastic collisions, and pressure exerted on container walls.

This page aims to simplify these concepts and highlight their real-world applications in thermodynamics, engineering, and atmospheric science.

Behaviour of Gas Molecules

Gas molecules move randomly in all directions and follow Newton's laws. The main assumptions of the kinetic theory are:



- A gas consists of a large number of tiny molecules in continuous motion.
- These molecules undergo elastic collisions with each other and with the container walls.
- The actual volume of gas molecules is negligible compared to the container volume.
- Intermolecular forces are absent, except during collisions.
- The average kinetic energy of the molecules is directly proportional to the temperature of the gas.

These assumptions explain why gases expand to fill their containers and exert pressure on the walls.

Ideal Gas Equation

The Ideal Gas Law relates the pressure, volume, temperature, and number of moles of a gas:

where:

- P = Pressure of the gas
- V = Volume of the gas
- n = Number of moles
- R = Universal gas constant ()
- T = Temperature in Kelvin

This equation applies to ideal gases, which perfectly follow the kinetic theory assumptions.

Kinetic Theory of an Ideal Gas

The pressure exerted by a gas on the walls of its container is derived from the molecular motion. According to kinetic theory:

— —

where:

- = Mass of a single gas molecule
- = Number of molecules in the gas
- = Volume of the gas
- = Root mean square (rms) speed of gas molecules

This equation shows that pressure is directly related to the average kinetic energy of the gas molecules, meaning that higher temperature results in higher pressure.

Kinetic Interpretation of Temperature

Temperature is a direct measure of the average kinetic energy of gas molecules. The relation between kinetic energy and temperature is:

—

where:

- = Average kinetic energy per molecule
- = Boltzmann's constant ()
- = Temperature in Kelvin

This equation explains that:

- As temperature increases, molecular motion becomes faster.
- As temperature decreases, molecular motion slows down.

This interpretation confirms that temperature determines the energy of molecules rather than the other way around.

Degrees of Freedom

The degree of freedom of a gas molecule is the number of independent ways it can store energy. The number of degrees of freedom depends on the type of molecular motion:

- Monatomic gases (e.g., He, Ar) $\rightarrow$ 3 degrees of freedom (motion in x, y, and z directions).
- Diatomic gases (e.g., O₂, N₂) $\rightarrow$ 5 degrees of freedom (3 translational + 2 rotational).
- Polyatomic gases (e.g., CO₂, CH₄) $\rightarrow$ 6 or more degrees of freedom (translational, rotational, and vibrational).

Degrees of freedom help in energy distribution calculations.

Law of Equipartition of Energy

The Law of Equipartition of Energy states that energy is equally distributed among all degrees of freedom. The energy per degree of freedom is:

—

The total average energy per molecule is:

—

For one mole of gas, the total kinetic energy is:

—

where f is the number of degrees of freedom.

For different gases:

- Monatomic gas ($f = 3$) → —
- Diatomic gas ($f = 5$) → —
- Polyatomic gas ($f = 6$ or more) → —

List of Important Formulas

S.No	Name of the Concept	Formula
1	Average Kinetic Energy	—
2	Average Pressure of Gas	—
3	Law of Equipartition of Energy	—
4	Most Probable Speed	— —
5	Average Speed	— —
6	Root Mean Square (RMS) Speed	— —
7	Mean Free Path	—

Conclusion

The Kinetic Theory of Gases provides a fundamental understanding of gas behaviour through molecular motion. It explains key gas laws and their role in thermodynamics, engineering, and atmospheric science. By grasping this theory, we better understand how gases interact, helping in advancements like engine efficiency, weather predictions, and industrial applications.

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